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AY2425-Sem1-Midterm-EE2211 Ans

Introduction to Machine Learning (National University of Singapore)

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# of Questions: 23
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Question #: 1

We would like to design a linear regression function to predict future cognitive performance and depressive severity in the elderly. We have collected two clinical features of 5 participants (x_1, x_2) that are related to future cognitive decline and depression. We also have the five-year follow-up data of cognitive scores (1st column of **Y**) and depressive scores (2nd column of **Y**) of these 5 participants (see below).

$$\mathbf{X} = \begin{bmatrix} 50 & 10 \\ 40 & 7 \\ 65 & 12 \\ 70 & 5 \\ 75 & 4 \end{bmatrix}, \mathbf{Y} = \begin{bmatrix} 9 & 3 \\ 6 & 7 \\ 5 & 6 \\ 3 & 1 \\ 2 & 9 \end{bmatrix}.$$

i) Perform least square estimation based on the existing training data. What is the mean of squared error of the estimated model for cognitive scores? __1__ (up to 4 decimal places) What is the mean of squared error of the estimated model for depressive scores?__2__ (up to 4 decimal places)

ii) If we have a new participant with the two clinical features (x_1, x_2) =(42, 8). What is his/her five-year follow-up cognitive score? __3__ (up to 4 decimal places). What is his/her five-year follow-up depressive score? __4__ (up to 4 decimal places).

iii) Say now instead of linear regression, we would like to use polynomial regression of order 3 for prediction of the same new participant with the two

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clinical features (x_1, x_2) = (42, 8). What is his/her five-year follow-up cognitive score? __5__ (up to 4 decimal places). What is his/her five-year follow-up depressive score? __6__ (up to 4 decimal places)

1. Range - **1.22 to 1.23**
2. Range - **8.01 to 8.02**
3. Range - **7.24 to 7.25**
4. Range - **5.23 to 5.24**
5. Range - **7.03 to 7.04**
6. Range - **5.71 to 5.72**

Question #: 2

The linear regression is the special case of polynomial regression of order 1.

✓A. True
B. False

Question #: 3

The solution for a ridge regression problem with feature vectors in **X** and targets in **y** can be written as

$$\hat{\mathbf{w}} = (\mathbf{X}^T\mathbf{X}+\lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

for λ >0. Which of the following is/are true?

I. As λ increases,

$$\hat{\mathbf{w}}^T\hat{\mathbf{w}}$$

increases.

II. As λ increases,

$$\hat{\mathbf{w}}^T\hat{\mathbf{w}}$$

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decreases.

III. As λ increases, there is more penalty on parameters.

IV. As λ increases, there is less penalty on parameters.

A. I &III
B. I &IV
C. II &III
D. II &IV
E. None of the above

Correct Answer: C

Question #: 4

The values of feature x and their corresponding values of target y are shown in the table below.

x	3	4	5	6	7
y	5	4.5	4	3.5	3

Based on the least square regression line y =a *x* +b, please select the correct option below.

I) The estimated a* and b* minimize the sum of squared errors between the predicted y' and the target output y for all training samples.

II) The sum of squared errors between the predicted y' and the target output y for all training samples is called the loss function.

III) The estimation of y at *x* =4.5 may be accurate as the prediction range is within the given data range.

IV) The estimation of y at *x* =1000 should be accurate as the prediction is linear.

✓A. I &III
B. III
C. I, II &III
D. I &IV
E. None of the above

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Question #: 5

Which of the following is/are true about matrix below?

$$\mathbf{A} = \begin{bmatrix} 1 & 4 & 2 \\ -3 & 10 & 5 \end{bmatrix}$$

I) **A** is invertible
II) **A** is left invertible
III) **A** is right invertible
IV) **A** has no determinant

A. I &III
✓B. III &IV
C. II &IV
D. II, III &IV
E. None of the above

Question #: 6

This question is related to determination of types of system where an appropriate solution can be found subsequently. The following matrix has a left inverse.

$$\mathbf{X} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

A. True
✓B. False

Question #: 7

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Which of the following statement(s) is/are true?

I. Given three samples of two-dimensional data points

$$\mathbf{X} = \begin{bmatrix} 1 & 4 \\ 5 & -1 \\ 2 & 3 \end{bmatrix}$$

with the corresponding target vector

$$\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

. Suppose you want to use a full second-order polynomial model to fit these data, the polynomials model has 6 parameters to learn.

II. Given three samples of two-dimensional data points

$$\mathbf{X} = \begin{bmatrix} 1 & 4 \\ 5 & -1 \\ 2 & 3 \end{bmatrix}$$

with the corresponding target vector

$$\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

. Suppose you want to use a full second-order polynomial model to fit these data, the polynomials model is an under-determined system.

III. Suppose we have 5 training samples, we would like to perform multi-class classification to predict y (4 classes) using input with 6 features using the optimal linear regression model with an offset term. In this case, the estimated parameter matrix **W** is of dimension 6 x 4.

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IV. Suppose we have 5 training samples, we would like to perform multi-class classification to predict y (4 classes) using input with 6 features using the optimal linear regression model with an offset term. In this case, the estimated parameter matrix **W** is of dimension 4 x 6.

A. I &II
B. I &III
C. I &IV
D. I
E. None of the others

Correct Answer: A

Question #: 8

A learning problem is attempted by minimizing the sum of squared errors given by

$$\sum_{i=1}^{m}(f_{\mathbf{w},b}(\mathbf{x}_i) - y_i)^2 = (\mathbf{X}\mathbf{w} - \mathbf{y})^T(\mathbf{X}\mathbf{w} - \mathbf{y}) \quad \text{where} \quad \mathbf{X} \in \mathcal{R}^{10\times 11} \quad \text{and} \quad \mathbf{w} \in \mathcal{R}^{11\times 1}$$

denote respectively the data matrix and the parameter vector. The outcome must be a unique solution since the square function is convex.

A. True
✓B. False

Question #: 9

Interval data is categorical, while ordinal data is numerical.

A. True
✓B. False

Question #: 10

Data imputation concerns handling missing feature values, and is a common practice in data visualization.

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A. True
✓B. False

Question #: 11

The Simpson's paradox phenomenon can be solved using Randomized Controlled Trial (RCT).

A. True
✓B. False

Question #: 12

Randomized Controlled Trial (RCT) is a way to conduct data visualization. Since we only need to assign participants into an experimental group and a control group, RCT is always feasible to conduct.

A. True
✓B. False

Question #: 13

A collection of vectors can be both linearly dependent and linearly independent at the same time.

A. True
✓B. False

Question #: 14

We are given the average daily temperature of Singapore in the past 20 years, and we are asked to learn a machine learning model that predicts the average temperature next Friday. This task, by nature, is a classification task.

A. True

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✓B. False

Question #: 15

Artificial intelligence is a subset of deductive inference, which adopts neural networks for making predictions.

A. True
✓B. False

Question #: 16

We have an unbiased die with six faces (i.e., 1,2,3,4,5,6), where each face has an equal probability of showing up in a rolling game.

Suppose we roll the die three times, and then count the sum of the faces showing up. For example, if the first time 1 shows up, the second time 3 shows up, the third time 2 shows up, the sum would be 1+3+2=6.

What is the probability of obtaining a sum being an **odd number smaller or equal to 5 (i.e., <=5)**?

A. 1/54

✓B. 7/216
C. 1/24
D. 1/216
E. 1/72
F. None of others is correct.

Question #: 17

Please select the correct option.

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(i) Data wrangling refers to the process of transforming and mapping data from one "raw" data form into another format with the intent of making it more appropriate and valuable for a variety of downstream purposes. Data wrangling applies to image data, but not audio data.

(ii) Without data imputation technique, the data records with missing values are completely useless and cannot be applied for training any machine learning model.

(iii) Nearest neighbour classifier works for categorical data only, but not numerical data.

(iv) Given two different sets of data samples, it is possible that these two sets end up having the same regression line, the same mean, and the same variance.

A. (i)
B. (ii)
C. (iii)
✓D. (iv)
E. (ii), (iv)
F. (i), (ii), (iv)
G. (ii), (iii)
H. None of the others.

Question #: 18

Please select the correct option.

(i) Data imputation applies to ratio data only, but not other types of data.

(ii) Data visualization applies to ratio data only, but not other types of data.

(iii) Data integrity applies to ratio data only, but not other types of data.

(iv) Binary coding applies to ratio data only, but not other types of data.

A. (i)
B. (ii)
C. (iii)
D. (iv)
E. (i), (ii), (iv)

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F. (i), (iv)
G. (i), (ii)
H. (ii), (iii), (iv)
✓I. None of others.

Question #: 19

We have a collection of 1,000 images from three class labels: cat, bird, and dog. With these images, we would like to train an image classifier that categorizes an input image into one of the three classes.

To ensure the 1,000 images are of good quality for training the classifier, we ask a well-trained human inspector to go through all the images, to label the images and remove noisy ones. Eventually, we revmoved 200 images of low quality suggested by the inspector, and use the remaining 800 images to train the classifier; the 800 images comprise 200 cat images, 300 bird images, and 300 dog images.

Jack and Mary are then asked to use these 800 images to derive a nearest neighbour classifier.

Jack extracts the following features from each image
1) the number of black pixels,
2) the number of white pixels,
3) the number of blue pixels.
As such, for each image, Jack obtains a 3x1 vector with non-negative integers.

Mary extracts the following features from each image
1) the number of black pixels,
2) the number of white pixels,
3) the number of green pixels,
4) the number of red pixels.
As such, for each image, Mary obtains a 4x1 vector with non-negative integers.

Please select the correct option.

A. The human inspection process can be considered as an unsupervised-learning step.

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B. Since Mary uses a 4x1 feature for each image and Jack uses a 3x1 feature, for sure Mary's classifier will lead to better classification results when compared to Jack's classifier.
C. The image labels (cat, bird, and dog) belong to ratio data.
D. There is no data wrangling involved in the above process.
E. (a), (b)
F. (a), (b), (d)
G. (b), (d)
✓H. None of the others.

Question #: 20

There are 300 students in a training camp, who are divided into two training groups: 200 students in Group A, 100 students in Group B.

Students in Group A are further divided to Team A-1, and Team A-2. There are 120 students in Team A-1, and 80 in Team A-2.
Students in Group B are further divided to Team B-1, and Team B-2. There are 50 students in Team B-1, and 50 in Team B-2.

The students take part in a fitness test.

Let G be the variable that stands for the group that a student belongs to, which can be 'a' (Group A), or 'b' (Group B).
Let T be the variable that stands for the team that a student belongs to, which can be 'a1' (Team A-1), 'a2' (Team A-2), 'b1' (Team B-1), or 'b2' (Team B-2).

Let X be the variable that stands for the result of the fitness test, which can be either 'p' (passing the test) or 'f' (failing the test).

In Group A, 150 students pass the test, among whom 100 are from Team A-1.
In Group B, 70 students pass the test, among whom 40 are from Team B-1.

What is the probability that **a student in this training camp passes the fitness test**, i.e., P(X='p')?

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Your answer here: 1 (Please write your answer to 3 decimal places, between 0 and 1).

What is the probability that **a student is from Team A-1 given that we know he/she is from Group A,** i.e., P(T='a1' | G='a')?

Your answer here: 2 (Please write your answer to 3 decimal places, between 0 and 1).

What is the probability that **a student is from Team B-1 given that we know he/she fails the test**, i.e., P(T='b1' | X='f')?

Your answer here: 3 (Please write your answer to 3 decimal places, between 0 and 1).

Which team has the **lowest rate of passing the test**?

Your answer here: 4 (Please choose one of the teams).

1. Range - Min:0.72 Max:0.74
2. Range - Min:0.59 Max:0.61
3. Range - Min:0.12 Max:0.13
4. Choice of: A-1 | A-2 | B-1 | B-2 - Correct Answer:B-2

Question #: 21

A vector function has the mapping of

$$f(\mathbf{x}): \mathcal{R}^3 \rightarrow \mathcal{R}^2$$

. The gradient of this function with respect to **x** is a matrix of partial derivatives, which is a 2×3 matrix.

✓A. True
B. False

Question #: 22

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This question is related to understanding of modelling assumptions.

$$f(\mathbf{x}) = 6x_1 + 9x_2 - 5x_3 + 3$$

is a linear function.

A. True
✓B. False

Question #: 23

Which of the following is/are true regarding derivatives?

I) Suppose g(**x**) is a *scalar* function of *b* variables where **x** is a *bx*1 vector. Without taking data points into consideration, the outcome of differentiation of g(**x**) w.r.t. **x** is a scalar.

II) Suppose f(**x**) is a *scalar* function of *d* variables where **x** is a *dx*1 vector. Without taking data points into consideration, the outcome of differentiation of f(**x**) w.r.t. **x** is a *dx1* vector.

III) If **f**(**x**) is a vector function of *size b*x1 and **x** is a *d*x1 vector, then differentiation of **f**(**x**) w.r.t. **x** is a *bxd* matrix.

IV) If **f**(**x**) is a vector function of *size b*x1 and **x** is a *d*x1 vector, then differentiation of **f**(**x**) w.r.t. **x** is a *dx1* vector.

A. I &III
B. I &IV
C. II &IV
✓D. II &III
E. None of the above

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